6.2 Fine-tuning

- in earlier chapters, we discussed how to train models on the Fashion-MNIST training dataset, which has only 60000 images
- we also described ImageNet, the most widely used large-scale image dataset in academia, which has more than 10 million images and 1000 objects
- however, the size of the datasets that we usually encounter will be between those of the two datasets

6.2 Fine-tuning

- suppose that we want to recognize different types of chairs from images, and then recommend purchase links to users
- one possible method is to first identify 100 common chairs, take 1000 images of different angles for each chair, and then train a classification model on the collected image dataset
- although this chair dataset may be larger than the Fashion-MNIST dataset, the number of examples is still less than one-tenth of that in ImageNet
- this may lead to overfitting of complicated models that are suitable for ImageNet, on this chair dataset
- besides, due to the limited amount of training examples, the accuracy of the trained model may not meet practical requirements

6.2 Fine-tuning

- in order to address the above problems, an obvious solution is to collect more data
- however, collecting and labeling data can take a lot of time and money
- for example, in order to collect the ImageNet dataset, researchers have spent millions of dollars from research funding
- although the current data collection cost has been significantly reduced, this cost still cannot be ignored

6.2 Fine-tuning

- another solution is to apply *transfer learning* to transfer the knowledge learned from the *source dataset* to the *target dataset*
- for example, although most of the images in the ImageNet dataset have nothing to do with chairs, the model trained on this dataset may extract more general image features, which can help identify edges, textures, shapes, and object composition
- these similar features may also be effective for recognizing chairs

6.2 Fine-tuning

- in this section, we will introduce a common technique in transfer learning: *fine-tuning*
- as shown in Figure 1, fine-tuning consists of the following four steps:
  1. Pretrain a neural network model, i.e., the *source model*, on a source dataset (e.g., the ImageNet dataset).
  2. Create a new neural network model, i.e., the *target model*. This copies all model designs and their parameters of the source model, except the output layer. We assume that these model parameters contain the knowledge learned from the source dataset, and this knowledge will also be applicable to the target dataset. We also assume that the output layer of the source model is closely related to the labels of the source dataset; thus, it is not used in the target model.
  3. Add an output layer to the target model, whose number of outputs is the number of categories in the target dataset. Then randomly initialize the model parameters of this layer.
  4. Train the target model on the target dataset, such as a chair dataset. The output layer will be trained from scratch, while the parameters of all the other layers are fine-tuned based on the parameters of the source model.
- when target datasets are much smaller than source datasets, fine-tuning helps to improve models' generalization ability

6.2 Fine-tuning

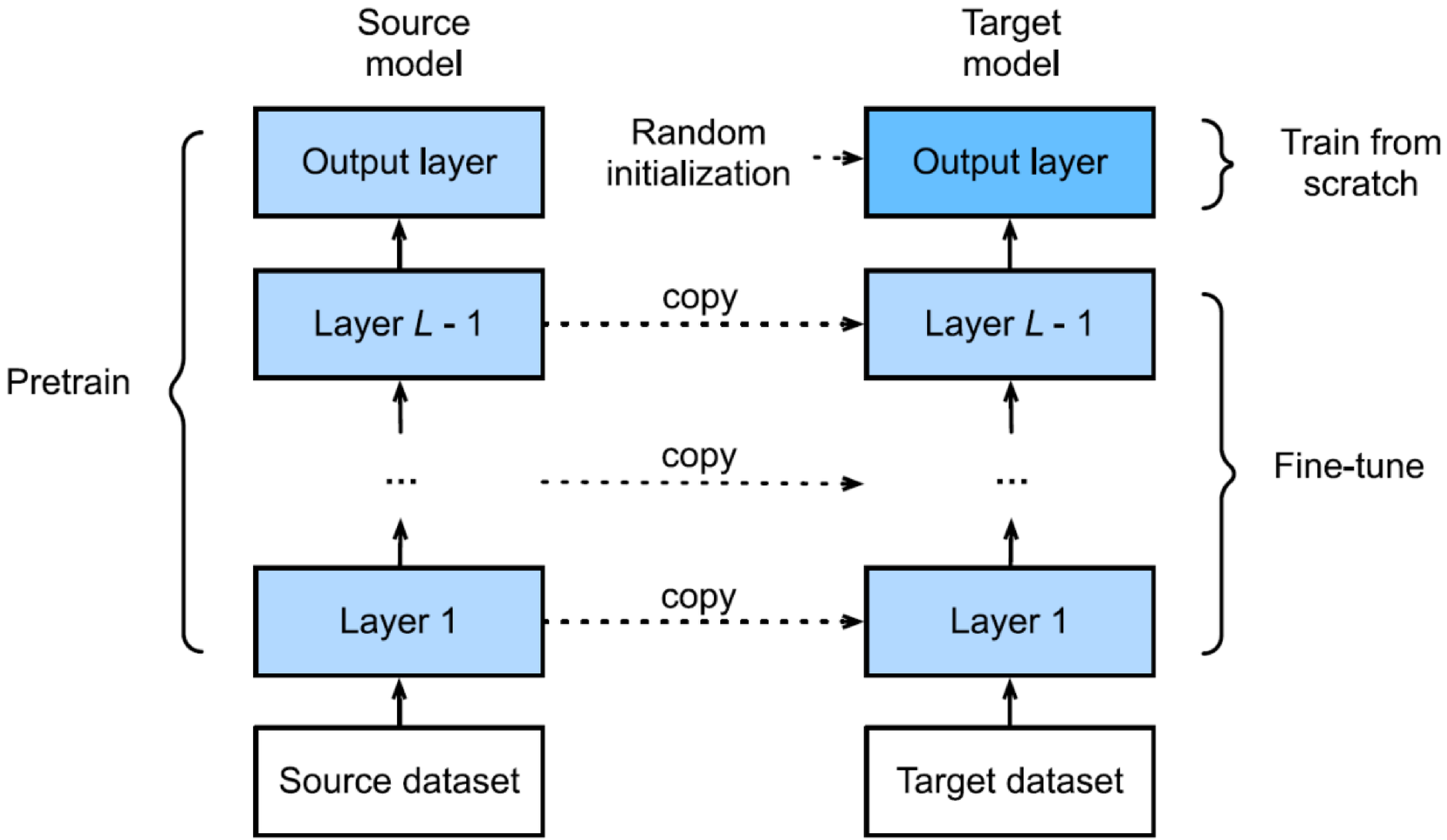


Figure 1: Fine-tuning.